Exam: Sheet 3
Level:
Course Title: Intelligent Information Systems

Course Code:
Date:
Time:

Q1 Briefly, explain the EMRs (Electronic Medical Records).

Answer:

- The EMR provides a systematized, digital collection of patient health information, which can be shared across healthcare settings. This includes notes and information collected by the clinicians in the office, clinic or hospital and contains the patient's medical history, diagnoses, and treatment plans. One can imagine that for each patient a substantial amount of healthcare data is accumulated over the course of their life, which can potentially be used to obtain a better understanding of medical conditions, diagnoses, and treatments.
- With the increasing adoption of EMRs and health monitoring technology, it is now possible to collect data together for individual patients and unite these to identify the needs of a subject to work toward a more unified goal for the entire population. Although there is a complex transition toward such consolidation from the "old"/established system of data logging to the new digitalized world, this is believed to lead to high value for patients along the line.

Q2 Briefly, explain important characteristics of new models in healthcare.

Answer:

- **Increased process management: Standardization of operational and clinical processes, increased use of new technologies and analytics including self-management and remote monitoring of patients, and more focus on performance.**
- **Focus on people: Strong leadership and effective people development and improved workforce models such as optimum use of highly qualified clinicians and displacement of less-skilled tasks to new professions such as health coaches.**
- **Focus on patients: Motivating patients to take an active role with self-management and larger differentiation of healthcare services depending on patient needs and desired outcomes.**
- **Wider scale: Increasing operating scale to support expansion of healthcare services and ensure improved performance management.**

Q3 Briefly, explain big data concept.

Answer:

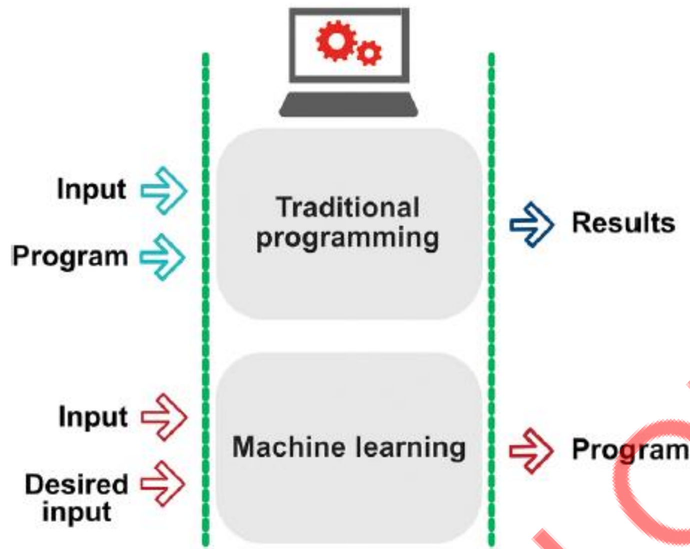
- **Big data is characterized by its high variety in the type of data and data source utilized as well as its applications. Especially within the healthcare sector, a large variety of data is acquired including new types of what did not exist prior to the emergence of smart devices. Indeed, when we consider data, variety may be as important a feature as the volume. This is due to the integration of many sources of data for better understanding and ultimately making sense of all the input . These sources of data can be collected in a variety of formats, including text, numbers, diagrams/graphs, video, audio, and a mixture of these. Examples of these data formats and sources include Images (medical imaging data), unstructured text (narratives, physician notes, reports, summaries), streaming (monitors, implants, wearables, smartphones), social media (Facebook and Twitter), structured data (NLP annotations, EMR data sources), and dark data (server logs, account information, emails, documents) .**

- **Data can generally be acquired in the format that is structured, semistructured, or unstructured and is acquired from either primary (EMRs, clinical decision support systems, etc.) or secondary sources (insurance companies, pharmacies, laboratories, etc.). One of the major complexities of big data is that it is mostly unstructured qualitative or incongruent. This makes it difficult to organize and process the data. Data preprocessing via text mining, image analysis, or other forms of processing is required to give structure to the data and extract relevant information from the raw data.**

Q4 Briefly, explain Machine learning concept. Illustrate your answer with a diagram.

Answer:

- **Machine learning (ML) is a subdiscipline of AI and constitutes a group of techniques used to solve data analysis by finding patterns among the data. These patterns provide the opportunity for understanding complex health situations (e.g., identify disease risk factors) or predicting future health outcomes (e.g., predict disease prognosis). ML draws competences from various areas of science including data science, statistics, and optimization. This tool is different from traditional programming in that it uses learning algorithms as opposed to traditional algorithms, and it utilizes mathematical models driven by probability and statistics and invokes predictions using available data to extrapolate unavailable data (Figure 1.6).**
- **Learning algorithms can learn from data and by feeding this (input) together with the desired results we can obtain the learning algorithm.**
- **ML can be classified into three learning types: (1) supervised learning, (2) unsupervised learning, and (3) semisupervised learning.**



The distinction between traditional programming and machine learning

Q5 Briefly, explain the types of artificial intelligence.

Answer:

- **Supervised learning uses datasets with prelabeled outcomes to train algorithms in solving classification and regression problems. It uses input variables to predict a defined outcome and is well-suited for regression problems, but it is demanding in time due to manual data labeling (supervision) and requires a large amount of data. The most common types of supervised learning algorithms include decision tree, Naïve Bayes, and Support Vector Machine. Decision trees group attributes by value-based sorting and is used for data classification purposes. Naïve Bayes is mainly used for classifying text data and employs methods based on probability of events. Support Vector Machine is also used for classification of data and uses principle of margin calculation, a way of assessing the distance between classes.**

- **Unsupervised learning uses datasets with unlabeled inputs and lets the algorithms extract patterns, features, and structure from the data, autonomously. It uses previously learned features to recognize data it is introduced to. It is used for detecting hidden patterns in the data and the objective is to find this pattern without human feedback. Unsupervised learning is useful for feature extraction and clustering. Common algorithms used include K-Means Clustering and principal component analysis (PCA). K-Means Clustering involves the automatic grouping of data into clusters (k distinct clusters) based on shared characteristics.**

- **Semisupervised learning is a mixture of supervised and unsupervised learning where only a subset of the data output is labeled, as fully labeled data is research intensive to acquire. It can be useful when unlabeled data is already available and labeling the data is tedious. Semisupervised learning methods include Generative Models, Self-Training, and Transductive**

Support Vector Machine. Generative models use mixed distributions such as Gaussian mixture models as identifiers for prediction. Self-training is a model where classifiers are first trained with labeled data and subsequently fed with unlabeled data and predictions are then added in the training set.

Q6 Briefly, explain ANNs (Artificial Neural Networks) and NLP (natural language processing).

Answer:

- **ANNs(Artificial Neural Networks) are models based on the brain and biological neurons and are used to perform computational problems. Similar to biological neurons, artificial neurons receive inputs from other neurons and each input carries a weight, which determines the importance of the input. Collectively, the artificial neurons comprise an ANN with numerous connections between neurons. One can picture a neural network as an extension of linear regression to capture complex nonlinear relationships between input variables and an outcome. ANNs can be trained using both supervised and unsupervised learning in the process of optimizing its prediction accuracy.**

- **NLP(natural language processing) is a technology that enables computers to understand and process human language. The process of reading and understanding language is complex as most languages do not follow logical and consistent rules and can be used in various ways. NLP leverages a selection of specialized ML tools to characterize textual information by building a pipeline, whereby the problem is broken up into smaller pieces that can be solved separately. In the healthcare space, it is used to extract structured data from unstructured clinical text data that is often hard for clinicians or professionals to process.**

- **NLP is regarded as one of the most developed of the healthcare analytics applications. It has demonstrated that it can bring value by extracting data from various text-based sources including medical records, clinical notes, and social media. For instance, a large part of clinical information is in the form of narrative text from physical examinations, lab reports and discharge summaries. It is important to note that these are** unstructured and incomprehensible to computer programs. Here, NLP is useful in extracting the relevant information from the narrative.

Q7 Briefly, explain CNNs and RNNs. And benefits of it in healthcare field.

Answer:

- **CNNs** typically have numbered nodes with a special connectivity pattern between its neurons based on the numbers and are especially good at dealing with unstructured and sequential data. They play an important role in image/audio/video recognition as well as NLP. The term convolution stems from convolution layer, a linear transformation that preserves spatial information in its dimensionality.
- **RNNs** are sequence-based and play an important role in NLP(Natural Language Processing), video processing, and other processes. These networks use the sequential information and perform the same recurrent task for each element of the sequence and capture the output information in their internal memory.
- **Reinforcement neural networks** are based on trial and error, much like learning in humans. This method differs from supervised learning in that inputs and outputs are not initially introduced.

At its core, all the above technology are here to serve a single purpose: Increase the quality of life for humans. From conditions that affect multiple organs, infections, and unwanted side effects of medications, the ability of healthcare workers to precisely pinpoint and extrapolate the trajectory of a condition will lead to significantly better outcomes for both the healthcare system and ultimately the patient.

It is worth stating that much of healthcare around the globe will have to deal with increasing complexities such as an increase in diversity in the clinical measurements, heterogeneity in clinical phenotypes, or the lack of specific biomarkers for often fatal conditions.

We believe that the implementation of the right policies that encompasses the usage of these tools can reduce the burden on the healthcare system and help thousands with new, high precision solutions.

Q8 Briefly, explain each of Complex algorithms, Digital health applications, and Omics-based tests.

Answer:

- **Complex algorithms: Machine learning algorithms are used with large datasets such as genetic information, demographic data, or electronic health records to provide prediction of prognosis and optimal treatment strategy.**
- **Digital health applications: Healthcare apps record and process data added by patients such as food intake, emotional state or activity, and health monitoring data from wearables, mobile sensors, and the likes. Some of these apps fall under precision medicine and use machine learning algorithms to find trends in the data and make better predictions and give personalized treatment advice.**
- **Omics-based tests: Genetic information from a population pool is used with machine learning algorithms to find correlations and predict treatment responses for the individual patient. In addition to genetic information, other biomarkers such as protein expression, gut microbiome, and metabolic profile are also employed with machine learning to enable personalized treatments.**

With my best wishes

Prof. Dr. Mohamed Okasha